Smart Infusion Pump Software Simulator

Software Requirements Specification (SRS)

Version 4.0 | Extended Edition with UI/UX Role Separation

Document Type	Software Requirements Specification
Version	4.0
Status	Extended – Role-Based UI Design
Target Platform	Mobile Application (iOS / Android)
Audience	Healthcare Educators, Students, IT Staff

1. Introduction & Scope

1.1 Overall System

The Smart Infusion Pump Software Simulator is an educational and training mobile application designed to replicate the behaviour of a modern infusion pump. It provides a realistic, risk-free environment for healthcare professionals and students to learn safe programming, alarm management, and clinical decision-making.

The simulator incorporates a layered architecture consisting of:

- A real-time simulation engine with event-driven state machine
- A configurable drug library with tiered safety limits
- A role-based access control (RBAC) system with three distinct user roles
- Comprehensive immutable audit logging and reporting tools
- A dedicated mobile UI per role to eliminate visual clutter and access confusion

1.2 Role Architecture Overview

The system enforces strict separation of duties across three roles. Each role has its own dedicated screens, navigation flow, and data access scope. No screen is shared between roles unless explicitly noted.

DOCTOR	Clinical governance: defines drug library, sets safety limits, reviews logs, generates reports. Does NOT operate the pump directly.

NURSE	Primary pump operator: selects drugs, programs infusion parameters, monitors real-time status, responds to alarms.

ADMIN	Technical operator: manages user accounts, maintains platform integrity.

1.3 Mobile Platform Constraints

As a mobile application, the following design principles apply across all roles:

- Single active session per device; no concurrent role switching without logout
- Large touch targets (minimum 44x44pt) for clinical simulation environment
- High-contrast colour scheme; WCAG AA compliance minimum
- Offline-capable simulation engine; server sync for logs and reports
- Session timeout: 15 minutes for Doctor/Nurse; 10 minutes for Admin

2. Functional Requirements

2.1 Doctor Role — Functional Requirements

The Doctor role focuses entirely on clinical governance. This role NEVER sees the pump operation screen, alarm controls, or infusion parameters during active simulation. All Doctor functionality is accessed via a separate, data-oriented interface.

ID	Priority	Requirement Description
F-D-01	High	The system shall allow the Doctor to view, add, edit, and delete drugs in the drug library via a dedicated Drug Management screen.
F-D-02	Critical	For each drug, the Doctor shall define: standard concentration (mg/mL), default infusion rate (mL/hr), hard limits (system-enforced ceiling), and soft limits (warning threshold). All four fields are mandatory.
F-D-03	High	The Doctor shall have read-only access to all infusion logs(except other users logs), including timestamps, events, alarms, and dose changes. No edit or delete access to logs.
F-D-04	Medium	The Doctor shall be able to generate reports that includes : (a) Infusion Summary, (b) Alarm Frequency. Be able to export as a PDF file

2.2 Nurse Role — Functional Requirements

The Nurse role is the primary pump operator interface. The UI is designed to mimic real infusion pump operation with large controls, clear status indicators, and prioritised alarm visibility. The Nurse has NO access to drug library management, user administration, or system configuration.

ID	Priority	Requirement Description
F-N-01	Critical	The system shall allow the Nurse to select a drug from the predefined drug library. Only active (non-archived) drugs shall appear in the selection list.
F-N-02	Critical	The system shall allow the nurse to enter the patient's username during infusion session setup.

ID	Priority	Requirement Description
F-N-03	Critical	The Nurse shall be able to enter infusion parameters: concentration (if there is update from doctor), infusion rate (mL/hr), total volume to be infused (mL).
F-N-04	Critical	The system shall validate all inputs against hard and soft limits in real-time: hard limit violation = input rejected with error modal; soft limit violation = yellow warning banner requiring acknowledgement before proceeding.
F-N-05	Critical	The Nurse shall be able to Start, Pause, Resume an infusion via clearly labelled, colour-coded buttons on the main pump screen.
F-N-06	Critical	The Nurse shall have an Emergency Stop button (red, full-width, always visible) that immediately halts the infusion, sets pump to safe state.
F-N-07	High	The dashboard shall display real-time data refreshed every second: current pump state, infusion rate (mL/hr), volume infused (mL), volume remaining (mL), estimated time remaining, and battery level (%).
F-N-08	Critical	The Nurse shall be able to view and acknowledge alarms via an Alarm Panel. Non-critical alarms: acknowledge to silence. Critical alarms: require resolution action before infusion can resume.
F-N-09	Low	The Nurse shall be able to trigger manual training alarms (for self-directed practice).
F-N-10	Medium	The Nurse shall be able to view a log preview panel showing the current session.

2.3 Admin Role — Functional Requirements

The Admin role manages the technical infrastructure of the simulator. The Admin interface is configuration-focused with no clinical content. Admins cannot operate the pump, access patient data, or modify drug limits directly (drug limits are the Doctor's domain).

ID	Priority	Requirement Description
F-A-01	Critical	The Admin shall be able to create, edit, and deactivate user accounts, assigning exactly one role per account (Nurse, Doctor, Admin or patient). No multi-role assignment.
F-A-02	Medium	The Admin shall be able to view system health metrics: memory usage, uptime, active sessions, and simulation engine status, Audit logs.

3. Non-Functional Requirements

3.1 Performance

ID	Priority	Requirement Description
N-D-01	High	Doctor report generation shall complete within 3 seconds for datasets up to 1,000 log entries on target mobile hardware.
N-N-01	Critical	The real-time pump dashboard shall refresh at a minimum of 1Hz (once per second) with UI update latency not exceeding 100ms.
N-N-02	Critical	The pump state machine shall respond to Nurse inputs (Start, Pause, Stop) within 500ms.
N-N-03	High	Logs related to nurse actions shall be updated within a maximum of 500 ms
N-A-01	Medium	Log export (up to 100,000 entries) shall complete within 10 seconds with progress indicator shown to Admin.

3.2 Usability

ID	Priority	Requirement Description
N-D-02	High	The Doctor UI shall require no more than 3 taps to reach any drug library function from the main Doctor dashboard.
N-D-03	High	The drug library editor shall validate all inputs in real-time (no negative limits, no concentration out of range) with inline error messages adjacent to the offending field.
N-N-04	High	The Nurse pump UI shall use high-contrast colours (minimum 4.5:1 ratio), and unambiguous iconography suitable for a clinical simulation environment.

4. Security Requirements

ID	Priority	Requirement Description
S-ALL-01	Critical	All users must authenticate with a unique user ID and password before accessing any role-specific screen
S-D-01	Critical	RBAC enforcement: only Doctor-role accounts may access drug library editing and report generation.
S-D-02	High	All Doctor actions (add/edit/delete drug, generate report) shall be logged with user ID, timestamp, and changed values in an immutable audit log.
S-N-01	Critical	The Nurse role is restricted from: drug library editing, user management, system configuration, and audit log viewing.
S-N-02	High	All critical Nurse actions (Start, Stop, Emergency Stop, alarm acknowledgement) shall be logged with the Nurse's user ID and session timestamp.
S-A-01	Critical	Only Admin-role accounts may access user management.
S-A-02	Critical	Admin account passwords shall be stored using bcrypt or Argon2 with a minimum cost factor of 12.
S-A-03	High	Admin session timeout is 10 minutes of inactivity (stricter than Doctor/Nurse at 15 minutes) due to elevated privilege level.

5. Safety Requirements

ID	Priority	Requirement Description
SF-D-01	Critical	Hard limits defined by the Doctor are enforced by the simulation engine. Any Nurse input exceeding a hard limit shall be rejected immediately with an alarm modal; infusion cannot start.
SF-D-02	High	Soft limit violations shall display a persistent yellow warning banner and require explicit acknowledgement (tap 'Acknowledge') before the Nurse may continue.
SF-D-03	Medium	The system shall prevent deletion of any drug that is referenced in an active infusion session.
SF-N-01	Critical	The infusion shall not start if any parameter violates a hard limit; the Start button shall be disabled and a blocking error displayed.
SF-N-02	Critical	The Emergency Stop button shall be prominently placed (bottom of screen, full-width, red), visible and tappable at all times during active infusion, and shall never be covered by modals.
SF-N-03	High	The system shall simulate battery drain and display a low battery warning at 20%. At 5% battery, infusion shall auto-stop and a critical alarm shall fire.
SF-A-01	Critical	The system shall prevent deletion of the last Admin account to avoid administrative lockout. A blocking error shall be shown.

6. UI/UX Design — Role Separation

This section defines the complete screen inventory for each role. Screens are strictly segregated: no Nurse screen is accessible from the Doctor flow, and vice versa. The Admin flow is entirely separate from clinical flows. This eliminates the role confusion identified in the design phase.

6.1 Authentication & Role Routing

All three roles share a single Login Screen. After successful authentication, the system reads the user's assigned role and routes to that role's dedicated navigation stack. There is no role selection at login — the role is pre-assigned by Admin.

Screen Name	Key UI Elements	Access Rule
Login Screen	User National ID, Password, Biometric login button, App logo	All roles — entry point only
Role Router	Invisible routing logic after login	System-level, no UI

6.2 Doctor Role — Screen Inventory

The Doctor interface uses a bottom navigation bar with 4 tabs. The colour theme is green-teal to visually distinguish from other roles. No pump operation controls appear anywhere in this flow.

Screen Name	Key UI Elements	Access Rule
Drug Library	Searchable list of drugs, Add Drug FAB, swipe-to-edit/delete.	Doctor only
Drug Editor	Form: drug name, concentration, default rate, hard limit, soft limit. Real-time validation.	Doctor only
Logs Viewer	Filterable log list (date, drug, patient). Read-only. Tap row for detail.	Doctor only
Reports	Report type selector, date filter, Preview pane, Export PDF button.	Doctor only

6.3 Nurse Role — Screen Inventory

The Nurse interface uses a simplified 3-tab bottom navigation. The colour theme is blue to reflect a clinical, focused environment. The Emergency Stop button is always rendered as a floating action element above the bottom navigation bar when an infusion is active.

Screen Name	Key UI Elements	Access Rule
Pump Dashboard	State indicator, Rate display, Volume infused/remaining, Time remaining, Battery bar, Progress bar. Central operational screen.	Nurse only
Drug Selection	Searchable drug list showing name and default rate. Tap to select and proceed.	Nurse only
Parameter Entry	Numeric inputs: concentration (if unlocked), rate, total volume. Live validation with inline hard/soft limit feedback.	Nurse only
Pump Controls	A single state-aware button dynamically switches between “Start” (green) and “Pause” (amber) based on the infusion state.	Nurse only
Emergency Stop	Full-width red button, always visible during active infusion. Requires single tap + hold 2s to prevent accidental trigger.	Nurse only — safety critical
Active Alarms	Displays currently triggered alarms with severity indicators and acknowledgment controls. Critical alarms remain locked until resolved.	Nurse only
Alarm History	Displays acknowledged and resolved alarms for monitoring and review purposes.	Nurse only
Session Log Preview	Events of current active session. Read-only. Tap for details.	Nurse only — read-only

6.4 Admin Role — Screen Inventory

The Admin interface uses a sidebar navigation (drawer) with 2 sections. The colour theme is purple-grey to visually distinguish from clinical roles. No clinical content (drugs, patient data, alarm panels) appears in this flow.

Screen Name	Key UI Elements	Access Rule
Admin Dashboard	System health cards: uptime, active users, active simulations. Alerts for pending issues.	Admin only
User Management	User list with role badge, Add User button, edit/deactivate actions. Role assignment dropdown in editor.	Admin only
Logs Viewer	Filterable log list (date, drug, patient). Read-only. Tap row for detail.	Admin only

6.5 Shared UI Rules (All Roles)

The following UI conventions apply across all role flows to maintain consistency:

- Role identity: a persistent role badge (coloured chip: green=Doctor, blue=Nurse, purple=Admin) is shown in the top navigation bar at all times
- Logout: accessible from top-right menu in all flows; prompts confirmation if simulation is active
- Session timeout warning: a 60-second countdown banner appears before session expiry with 'Stay logged in' option
- No cross-role navigation: deep links or back-navigation shall never route a user into another role's screen tree

Accessibility: all interactive elements have accessibility labels; minimum contrast ratio 4.5:1; dynamic type supporte

7. Verification Requirements

7.1 Doctor Verification

ID	Priority	Requirement Description
V-D-01	Test Case	Add a new drug with valid limits. Verify it appears in Drug Library list and is selectable by Nurse role.
V-D-02	Test Case	Attempt to set a hard limit of -5 mg/mL. System must reject with inline validation error and not save.
V-D-03	Test Case	Edit a drug. Open audit trail and verify entry shows: user ID, timestamp, field changed, old value, new value.
V-D-04	Test Case	Attempt to delete a drug currently in use in an active simulation. System must block deletion with explanation.
V-D-05	UI Test	Log in as Doctor. Verify no pump operation buttons, no alarm acknowledgement, and no user management options are accessible.

7.2 Nurse Verification

ID	Priority	Requirement Description
V-N-01	Test Case	Program a valid infusion. Start it. Verify volume infused counter increments and time remaining decreases in real-time.
V-N-02	Test Case	Enter an infusion rate above the hard limit. Verify Start button is disabled and error message is shown.
V-N-03	Test Case	Enter a rate above the soft limit. Verify yellow warning appears and requires acknowledgement before Start is enabled.
V-N-04	Test Case	Trigger an occlusion alarm mid-infusion. Verify pump auto-pauses, alarm panel shows critical alarm, and infusion cannot resume until alarm is resolved.
V-N-05	Test Case	Press Emergency Stop. Verify pump enters shutdown state within 500ms, and event is logged.
V-N-06	UI Test	Log in as Nurse. Verify no drug library editor, no user management, and no system configuration screens are accessible.

7.3 Admin Verification

ID	Priority	Requirement Description
V-A-01	Test Case	Create a new Nurse account. Log in with it and verify Nurse-only screens are accessible and Admin screens are not.
V-A-02	Test Case	Attempt to delete the last Admin account. System must block with error message.
V-A-03	Inspection	Review audit log after 5 Admin configuration changes. Verify all 5 entries are present with user ID, timestamp, and changed values.
V-A-04	UI Test	Log in as Admin. Verify no pump operation controls, no drug library editor.

8. Assumptions & Constraints

8.1 Assumptions

- All users are authenticated and assigned exactly one role. Multi-role accounts are not supported.
- The simulator operates exclusively in a non-clinical, training environment. No real patient data is processed.
- The drug library is initially populated by an authorised Doctor or Admin before training sessions begin.
- Target mobile hardware meets minimum performance specifications: 4GB RAM, 64-bit processor, iOS 15+ or Android 12+.
- Network connectivity is required for log synchronisation and report export; core simulation runs offline.

8.2 Constraints

- The simulator cannot fully replicate real-world sensor physics or mechanical pump failures.
- Simulation timing accuracy may degrade on low-performance or battery-constrained devices.
- Safety logic is simplified for educational purposes and must not be used for actual patient care.
- The system must comply with institutional data protection policies (AES-256 encryption for stored logs).
- Hard limits defined by the Doctor always take precedence over Admin global limits when both are set.

Appendix A: Role-Permission Matrix

The table below summarises all major system features and which roles may access them. This matrix resolves the role boundary ambiguities in the original design.

Feature / Screen	Doctor	Nurse	Admin
Drug Library (view)	✓	View-only	✗
Drug Library (edit/add/delete)	✓	✗	✗
Safety Limits (drug-specific)	✓	✗	✗
Pump Operation (Start/Pause)	✗	✓	✗
Emergency Stop	✗	✓	✗
Parameter Entry	✗	✓	✗
Alarm Acknowledgement - Resolving	✗	✓	✗
Real-time Pump Dashboard	✗	✓	✗
Infusion Logs (view)	✓	Session only	✓ (except other users login)
Reports & Analytics	✓	✗	✗
PDF Export	✓	✗	✗
User Management	✗	✗	✓

Legend:

- ✓ = Full access
- ✗ = No access (attempting to navigate to this feature results in an access-denied screen)
- View-only / Session only / Training only = Restricted read access as described in the adjacent cell